

Math Intellect

AI-Assisted Mathematics Learning Platform

Public GitHub edition | Functional MVP and 51-case technical validation

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Purpose and publication scope

This condensed report presents the implemented Math Intellect minimum viable product (MVP), its processing design, recorded technical validation, observed limitations and planned development. It is derived from the full Project Report V2.1 and is intended for public technical inspection alongside the repository README, sanitized workflow, sample cases and validation evidence.

It does not represent a production deployment, official Cambridge affiliation, independently verified marking equivalence, measured educational improvement, confirmed customers or commercial revenue.

Repository

<https://github.com/EduAILabs/math-intellect>

Detailed evidence paths are listed on the last page. This document intentionally omits personal career records, student-identifying information, credential values and private operational settings.

1. Project overview and current MVP

Math Intellect addresses the need for timely, structured mathematics support outside classroom hours. A learner submits an image of a mathematics question through a LINE Official Account. A locally hosted n8n workflow retrieves the image, coordinates AI interpretation and reasoning stages, prepares a worked solution with examiner-inspired assessment guidance and returns a formatted response through LINE when its processing checks permit delivery.

The present technical-validation scope is Cambridge IGCSE Mathematics 0580. Cambridge IGCSE Additional Mathematics 0606 is a curriculum-development reference, not a separately validated subject in the 51-case baseline.

Implemented technology

Component	Current function
LINE Official Account / Messaging API	Image intake and LINE reply
Locally hosted n8n	Webhook, image retrieval, orchestration and routing
OpenAI API	Vision and language processing
JavaScript	Response formatting
ngrok	Development tunnel to locally hosted webhook

Six-stage processing sequence

Question Analyzer → Solver → Examiner → Moderator → Quality Checker → Formatter.

Vision/image analysis precedes the six named stages. The Examiner produces examiner-inspired assessment guidance rather than an official Cambridge mark scheme. Moderation or quality-check failures can stop the normal reply route; a successful execution on one question does not establish reliability across question types.

Response structure

The LINE response is organised as Question Analysis & Assessment Guidance (header), Step-by-Step Solution (body) and Examiner-Inspired Assessment Guidance (tail). The sample cases and workflow documentation distinguish working behaviour from unsuccessful processing.

2. Technical validation – 51 recorded cases

The recorded validation covers 51 Cambridge IGCSE Mathematics 0580 test cases. Interpretation, answer accuracy, solution completeness, assessment alignment and LINE delivery are separate criteria with different applicability rules; conditional results must not be described as whole-dataset accuracy.

Recorded metric	Result
Test cases	51
Text interpretation	100.00 % (51/51)
Visual / diagram interpretation	4.17 % (1/24 applicable)
Overall question interpretation	54.90 % (28/51)
Final mathematical-answer accuracy	95.65 % (44/46 assessable)
Sub-question accuracy	76.62 %
Average solution completeness	77.78 %
Average assessment alignment	92.66 %
End-to-end LINE response success	90.20 % (46/51)

Interpretation of findings

The final-answer criterion was met in 44 of 46 assessable cases: 95.65% does not mean that 44 of all 51 submissions succeeded. Diagram interpretation passed in only 1 of 24 applicable cases, indicating a significant limitation in extracting information encoded in diagrams. LINE delivery occurred in 46 of 51 cases; delivery is not synonymous with mathematical correctness.

Completeness and assessment alignment are rubric-based technical measures, not independent confirmation of equivalence with official Cambridge marking. This baseline does not measure learning gains, time saved by teachers, adoption, willingness to pay or production-grade reliability.

3. Demonstration evidence and known limitations

Two contrasting samples

Sample	Observed behaviour
SAMPLE-01 – text-based question	The system produced a correct step-by-step answer and a visible LINE reply.
SAMPLE-02 – diagram-dependent question	Diagram givens were not extracted; Solver reported that it could not see the diagram; Moderator returned FAIL; Moderator Check routed to False Branch; no final LINE answer was delivered.

The two demonstrations illustrate behaviours of the MVP; they do not modify or replace the 51-case aggregate record.

Present limitations

- Locally hosted functional prototype; cloud production deployment and multi-user load validation are not complete.
- Diagram-information extraction is unreliable in the recorded baseline; a retry/request-for-clarification mechanism is planned.
- Persistent learner database, teacher dashboards, analytics, worksheet generation and email/PDF delivery are not implemented in the documented MVP.
- AI-generated solution and marking guidance require critical review; the platform is not affiliated with or certified by Cambridge International.
- No measured student learning impact, teacher time savings, paying customers, institutional partnerships or revenue are claimed.

Public workflow caveat

The repository workflow export is sanitized and inactive for public inspection. Credentials and environment-specific configuration must be supplied after import; successful execution of the sanitized copy has not been independently established. Sensitive access tokens must not be committed to the public repository.

4. Development direction and technical evidence

Proposed development, not deployed capabilities

Horizon	Planned focus
2026	Functional MVP validation and technical refinement
2027	Teacher productivity tools and cloud SaaS development
2028	Exploratory school partnerships and possible institutional licensing
2029 onward	Conditional regional expansion and AI learning analytics

Taiwan is a proposed base for product refinement and exploratory educator-led pilot discussions. Thailand, Japan, Singapore, Malaysia and Myanmar are possible research markets, not confirmed launch territories. Market priorities depend on local research, user needs, data protection, operating costs and pilot evidence.

Proposed commercialization

Potential revenue routes include B2C subscriptions for learners and teachers and B2B licensing for learning centres or schools if the relevant features are developed and validated. No pricing, customer demand, paying subscriptions, partnerships or revenue have been demonstrated. Pilots would evaluate usefulness, reliability, safeguarding and operating costs before commercial decisions.

Repository evidence index

Location	Purpose
README.md	Project scope, setup context and validation summary
workflow/	Sanitized n8n export and import/setup notes
diagram/	System architecture, reasoning pipeline and data flow
docs/	Technical, curriculum, QA, validation and planned business documentation
samples/	SAMPLE-01 success and SAMPLE-02 failure records
validation/Math_Intellect_Prototype_Validation_Record.xlsx	51 – cases scoring record
validation/Math_Intellect_Prototype_Validation_Evidence_Record.pdf	Validation summary
validation/Math_Intellect_Prototype_Validation_Evidence_Record.pplx	Illustrative validation evidence

Repository: <https://github.com/EduAILabs/math-intellect>

Evidence paths describe the intended published repository structure; ensure these files are present before linking this report publicly.